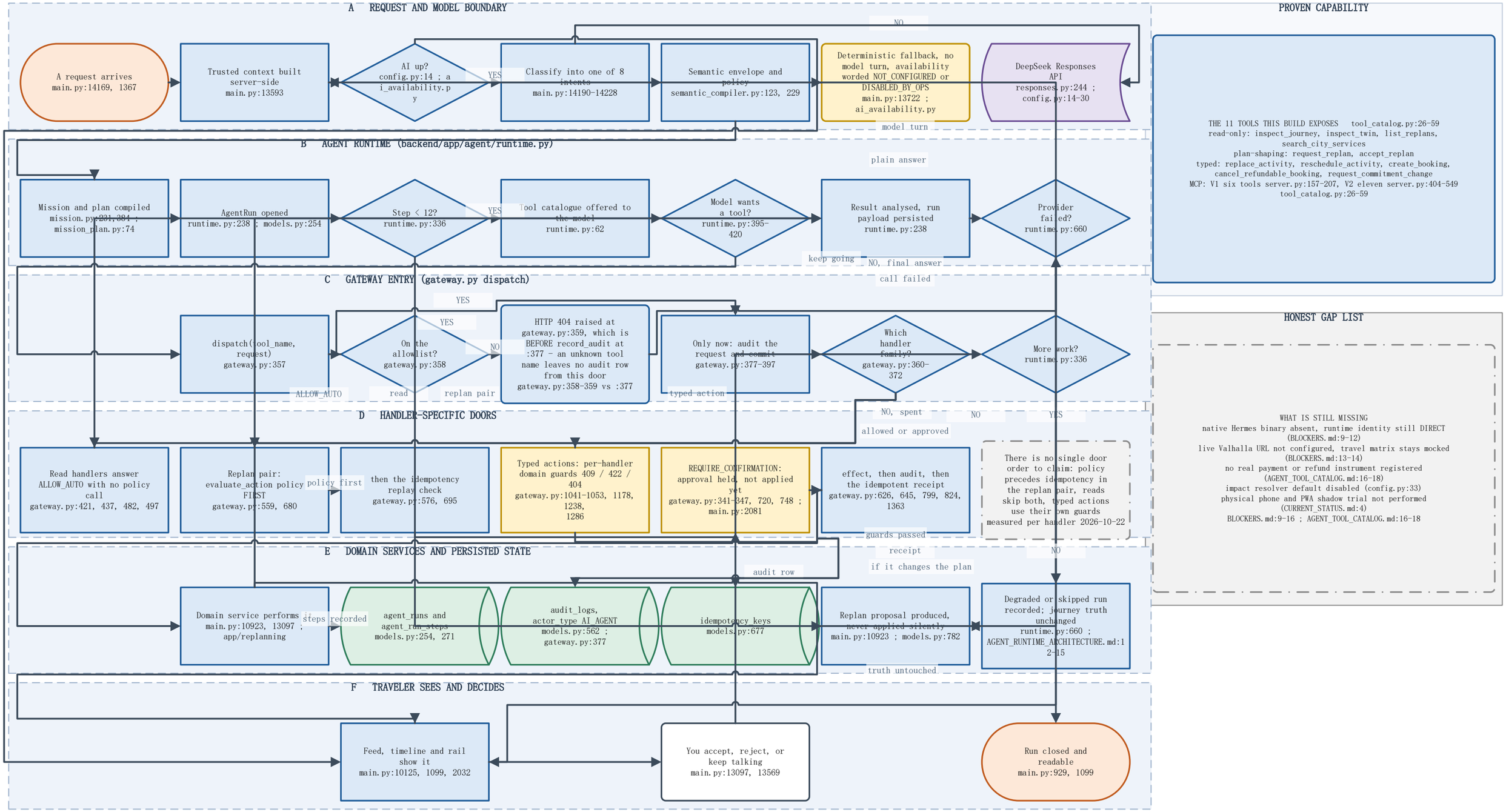
CityNexus | AI Agent Autonomy and Tool-Calling Workflow

CNX-SE-VIS-03 Agent workflow (measured handler paths)
Rev A - generated 2026-10-22 - source tree CityNexus @ review/boli-party-occupancy-scope-r1, github.com/Canshu-King/CityNexus

Six lanes, drawn against gateway.py as measured on 2026-10-22. The point is not that the agent is smart but that it is bounded: an unknown tool is refused before anything is written, reads answer without a policy call, the replan pair runs policy then idempotency, and typed actions carry their own domain guards. A failed provider becomes a recorded degraded run, never a quiet rewrite.



Legend

- Read in the source and cited line by line
- Present but not proven on the current line
- Missing or future work
- Agent-owned step
- Human in the loop
- Model boundary
- Persisted state

CORRECTION MADE IN THIS VERSION: an earlier draft drew one universal door order (allowlist, then idempotency, then authorization) and said an unknown tool was still audited. Neither is what the code does. dispatch() raises the 404 at gateway.py:359 and only reaches record_audit at :377 for a tool that passed the allowlist; and in request_replan the policy decision at :559 comes BEFORE the idempotency lookup at :576.

So the diagram now separates the three handler families, because they genuinely differ: read handlers return ALLOW_AUTO with no policy call (:421, :437, :482, :497); the replan pair runs policy then idempotency then effect then audit then receipt (:559, :576, :626, :645 and :680, :695, :799, :824); the typed actions raise their own domain guards (:1041-1053, :1178, :1238, :1286) and request_commitment_change audits at :1363.

max_steps=12 is a real bound at runtime.py:336; spending it is reported as a degraded run rather than retried invisibly.

The right-hand panels belong in the model. The coursework asks to separate what the code does today from what is planned, and claiming a closed automation loop here would be false.